

LÊ NAM KHÁNH

AI / ML ENGINEERING | MULTIMODAL RETRIEVAL | RESEARCH

Information Technology student | University of Science, VNU-HCM (HCMUS)

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AI CHALLENGE 2026	HCMUS CODING	CODEFORCES	SOICT 2026
Finalist Bảng A (Top Teams)	Champion 2026	Expert Max 1796	Paper First Author

PROFILE

Information Technology student at HCMUS with a rigorous competitive programming background and practical expertise in multimodal video retrieval and AI systems. First & corresponding author of SOICT 2026 research on structured temporal event alignment (GEMTRA) and Finalist at the Ho Chi Minh City AI Challenge 2026. Skilled in architecting end-to-end retrieval pipelines, fast algorithmic reasoning under constraints, and full-stack engineering.

EDUCATION

University of Science, VNU-HCM (HCMUS) Information Technology Second-year student	Current direction Multimodal video retrieval, temporal event reasoning, and foundation model systems.
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TECHNICAL SKILLS

Algorithms	Data structures, graph algorithms, dynamic programming, constraint optimization
Languages	Python, C++, TypeScript / JavaScript, Rust, SQL, Bash / PowerShell
AI / Retrieval	Visual embeddings (SigLIP2), FAISS, RF-DETR, OCR/ASR, VQA, GEMTRA alignment
Engineering	React 19 / Vite, Firebase / Firestore, Tauri, FastAPI, Git, testing, DRES API

ACHIEVEMENTS

AI Challenge HCMC 2026 Finalist (Bảng A) Team Reply 404	HCMUS Coding Challenge Champion 2026
SOICT 2026 Full Paper First & Corresponding Author (LNCS)	Codeforces Expert Max rating 1796
National Young Informatics First Prize (Central) Nat'l Honourable Mention	Provincial & Olympic 30/4 Provincial 2nd & 3rd Prizes Bronze Medal

RELEVANT STRENGTHS

- Fast at translating mathematical formulations & constraints into efficient implementations. - Resilient operator with high debugging agility under live competition countdowns. - Rigorous engineering mindset: separating empirical evidence from benchmark claims.	- Full-stack capability: bridging ML/AI backend systems with fast, intuitive user experiences. - Strong team collaborator across research ideation, system benchmarking, and code review. - English: IELTS 6.5 GitHub: lenamkhanhh Codeforces: Average2k7
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SELECTED PROJECTS

Reply 404 — Video Retrieval System Python SigLIP2 FAISS GEMTRA DP Alignment DRES API Engineered an interactive multimodal video search platform for 1,487 videos & 533K keyframes with sub-7ms temporal DP alignment (GEMTRA) and evidence-linked VQA under AIC countdowns. SOICT 2026 Paper GitHub / HCMAIC-Retrieval
TripFlow Workbench React 19 TypeScript Vite Firebase Auth & Firestore Rules Built a collaborative travel planning workbench featuring 4 realtime workflows: timeline scheduling with priority/assignees, expense splitting, and team member synchronization. Live App / mxhuit26.vercel.app GitHub
AI Account Switcher Tauri Rust React Local API Gateway CLI Wrappers Desktop app managing multi-account rotation for AI coding assistants (Claude Code, Codex) with an OpenAI-compatible local proxy gateway and real-time quota tracking. GitHub / ai-switcher
RESEARCH-ORIENTED INTERESTS Multimodal Video Retrieval Temporal Event Reasoning Vision-Language Models AI Agents Also building competitive programming training artifacts and reproducible benchmarks.